



Crisis or Not, the State of the Union Stays

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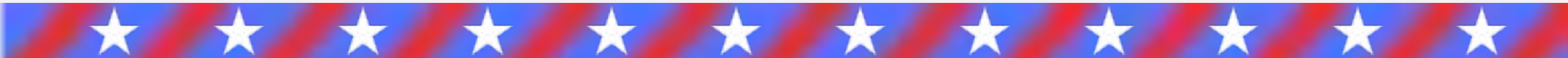
Optimistic

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Leader), Margot Brady, & Patrick
Yin

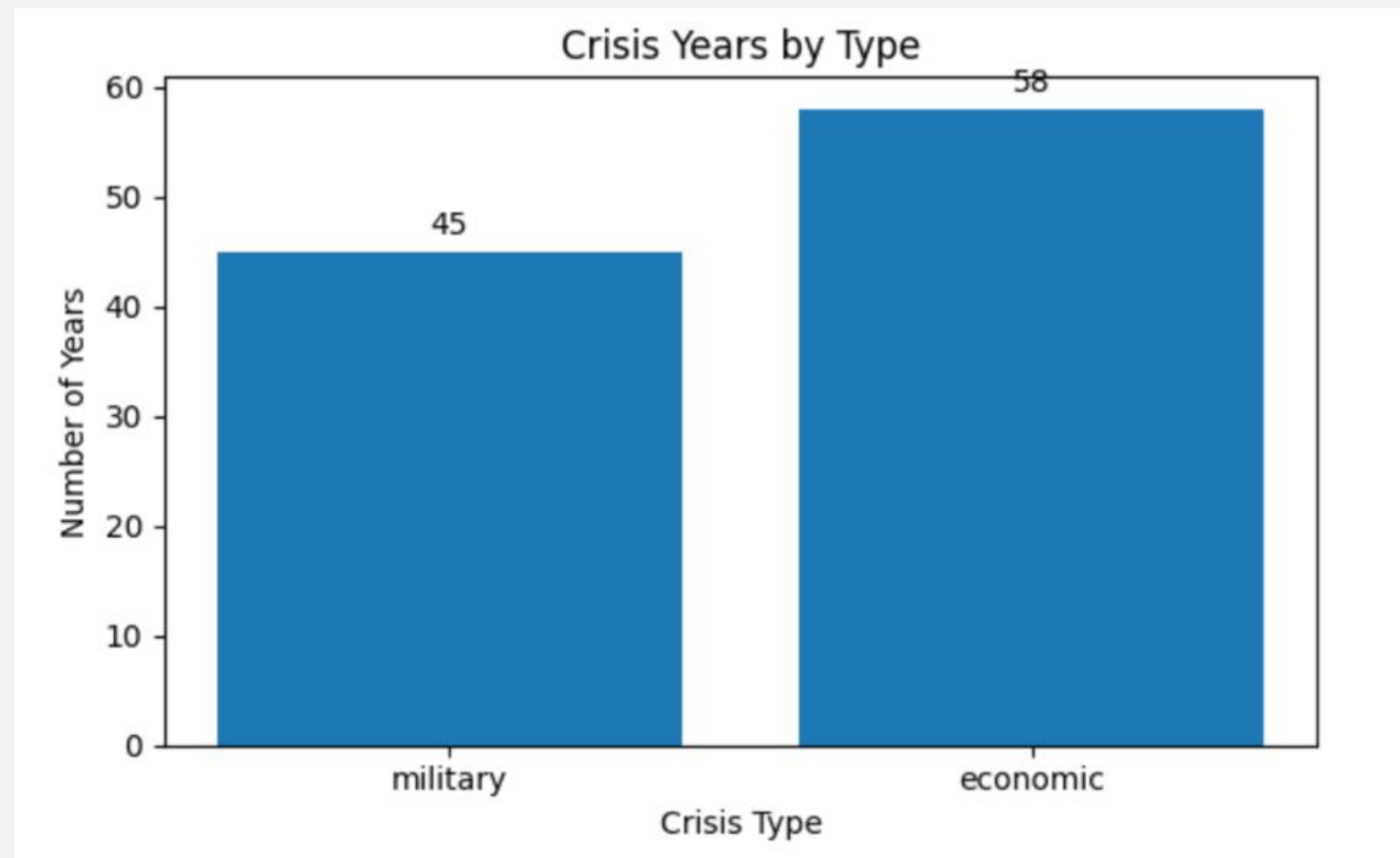
DS 4002
02/16/2026

Project Details

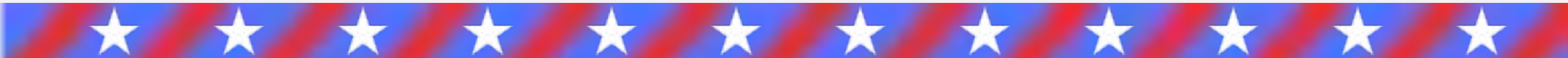
- **Motivation:**
 - To understand how presidential rhetoric functions as a leadership tool during challenging times
- **Hypothesis:** During times of national crisis, *more than 50%* of State of the Union addresses given in the same year as the crisis will be classified as more optimistic.
- **Research Question:** Will the sentiment score of State of the Union addresses be significantly lower during years classified as periods of national crisis compared to the average score during years of relative stability?
- **Modeling Approach:** Sentiment Analysis using Hugging Face Transformers library
- **Goal:** We will conduct a comparative sentiment analysis of every single United States State of the Union addresses *since 1850*



Defining “National Crisis”



Event	Start	End
Recession	August 1918	March 1919
Recession	January 1920	July 1921
Recession	October 1926	November 1927
Great Depression	August 1929	1939
Recession	May 1937	June 1938
World War II	September 1, 1939	September 2, 1945
Recession	November 1948	October 1949
Korean War	June 25, 1950	July 27, 1953
Recession	July 1953	May 1954
Vietnam War	November 1, 1955	April 30, 1975
Recession	August 1957	April 1958
Recession	April 1960	February 1961
Cuban Missile Crisis	October 1962	October 1962
Recession	November 1973	March 1975
Recession	January 1980	July 1980
Recession	July 1981	November 1982
Early 90s Recession	July 1990	March 1991
Dot Com Recession	March 2001	November 2001
War on Terror	September 2001	May 2003
Great Recession	December 2007	June 2009
COVID-19 Pandemic / Recession	March 2020	May 2020



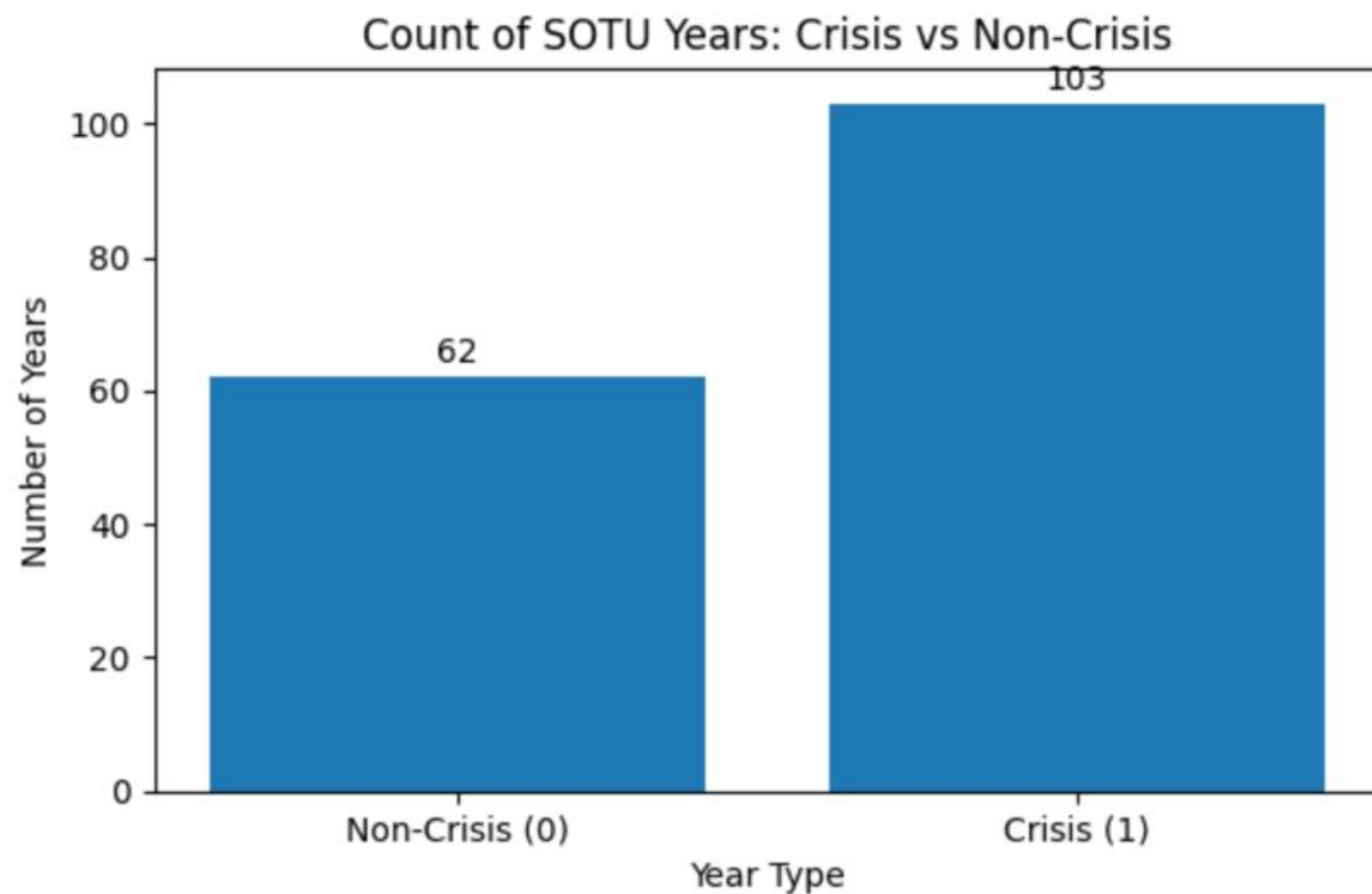
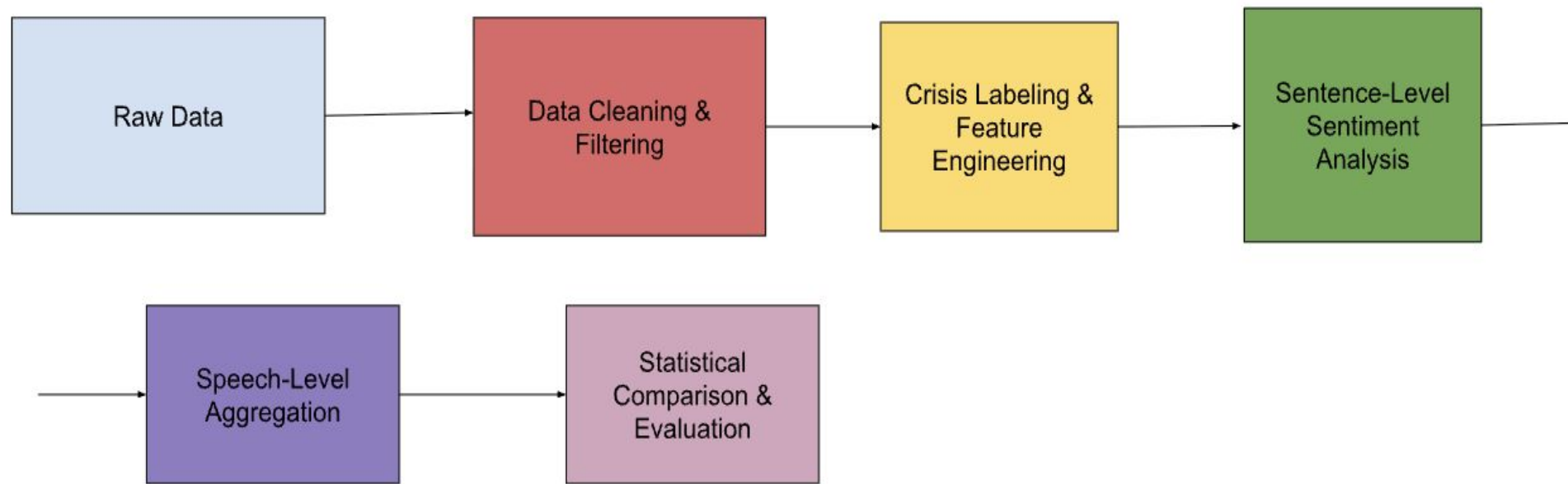
Data Acquisition

- Acquisition: Kaggle. There were no licensing or ethical concerns.
- Dataset size: 169 rows
- We added any missing data (post 2019 speeches), and then added two new columns to determine which years were in times of crisis
 - Times of crisis were determined from dates where the U.S. was in a recession, or where the U.S. was actively in war or a military crisis.
- We also removed years before 1850 due to missing data about economic recessions, and cleaned speech text for irrelevant sentences (audience reactions)

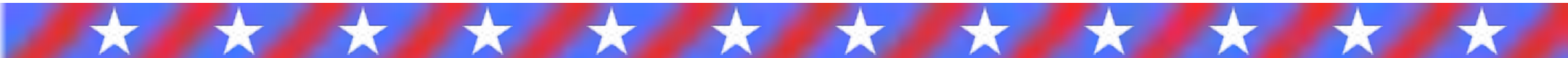
Data Dictionary:

Column	Description	Potential Responses
President	The President giving the State of the Union Address	Barack Obama
Year	The year the address was given	2008
Title	Which State of the Union address was it for that President	Barack Obama's First State of the Union Address
Text	Full text of the SOTU address	['Madame Speaker, Mr. Vice President, Members of Congress, and the First Lady of the United States:',.....
Is_Crisis	Whether the U.S. was facing an economic or military crisis (1) or not (0)	0, 1
Crisis_Name	Description of what crisis the country was facing at that time	The Great Recession

Analysis Plan & Justification



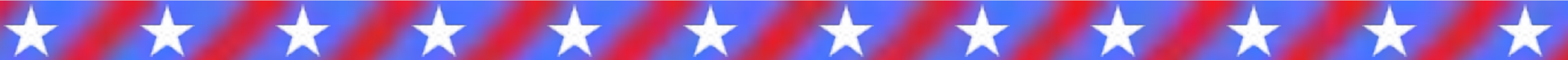
- After cleaning the dataset, we conducted **EDA** to examine the number of crisis vs. non-crisis years
 - Allowed us to examine if there was a class imbalance before testing
- A challenge we faced was Excel's character limit
 - We had to split a lot of the speeches into two separate rows
 - Had to figure out the right date to start with
- After totaling the number of positive and negative sentences in one speech, we compared the numbers of each and concluded the speech as optimistic or negative based on the results.
 - We also tracked if that speech was given in a crisis or not, and **tallied the results** to compare after all speeches were classified to see if a crisis had any relationship with the sentiment of the speech.
 - Determined threshold (>50% positive)



Tricky Analysis Decision

Defining a “Crisis Year”

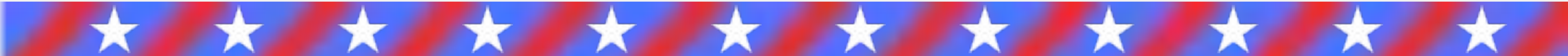
- Crisis periods **do not** have clear start and end boundaries, yet our model required a binary classification
- Real-world crises are continuous, but our analysis required event windows
- For example, while the War on Terror technically spanned nearly 20 years, domestic economic and political disruption was not constant throughout that period. Treating the entire span as a continuous crisis would overgeneralize its impact.
- Because, the definition of a crisis is “**subjective**”, we only limited to military and economic events and then used objective dates to verify periods of crisis.
- We operationalized crisis years using defined event windows and verified classifications with external historical sources
- Different labeling windows could meaningfully change our overall findings, so this step was extremely important, as well as very tricky.



Bias and Uncertainty Validation

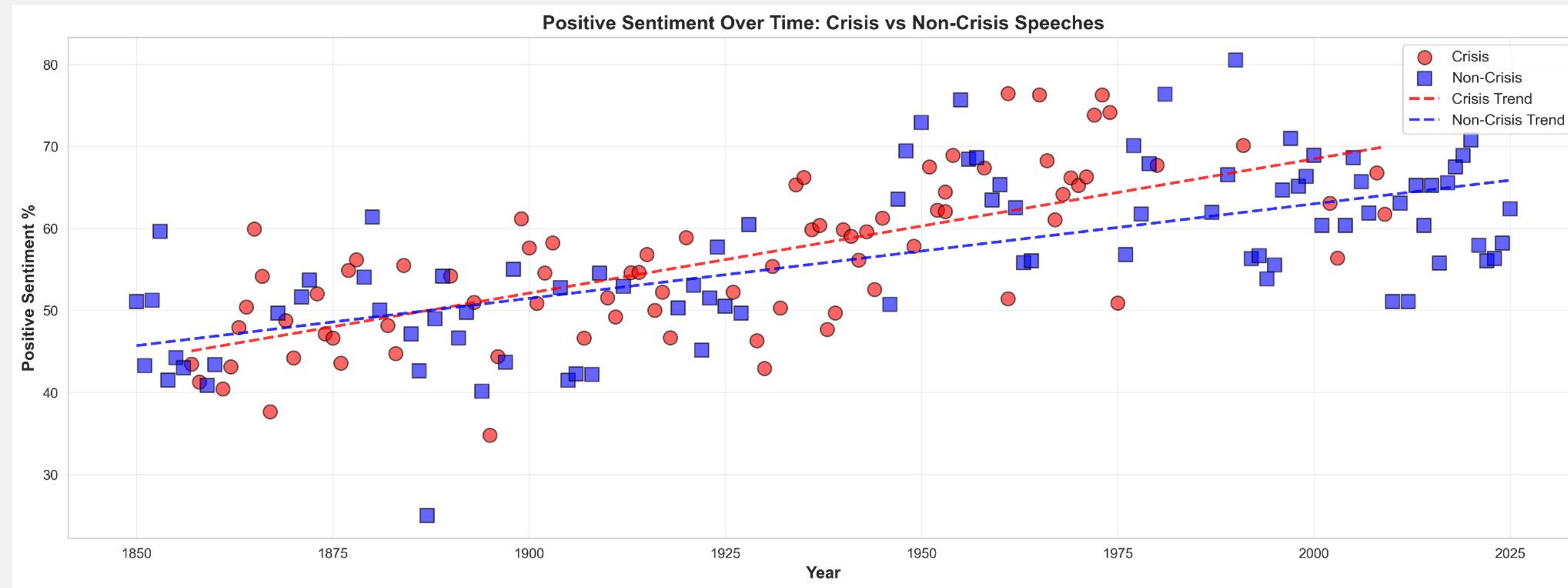
- The Date of The SOTU wasn't given at the same time every year in the past, so we had to go back and manually verify dates.
- We manually added the past 5 years of missing data (2020 - 2025), to ensure it was up to date
- We limited our analysis to speeches beginning in 1850 to reduce extreme linguistic variation from early American English, as the Hugging Face model is trained on modern English
- Using a >50% positive threshold creates a strict cutoff
- We included model confidence scores (most above 95%) to show the strength of the model's predictions.
- To **limit bias** in classification, we relied on established historical and economic sources

year	president	title	total_sentences	positive_count	negative_count	positive_percentage	negative_percentage	avg_positive_confidence	avg_negative_confidence	overall_sentiment	num_paragraphs	total_characters	is_crisis	crisis_name
2008	George W. Bush	George Walker Bush's Eighth State of the Union Address	316	211	105	66.77	33.23	0.9736	0.9381	POSITIVE	1	34803	1	The Great Recession
2009	Barack Obama	Barack Obama's First State of the Union Address	285	176	109	61.75	38.25	0.9705	0.9648	POSITIVE	1	33769	1	The Great Recession

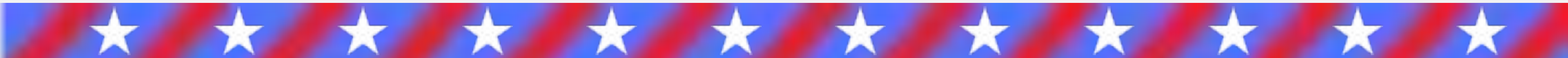


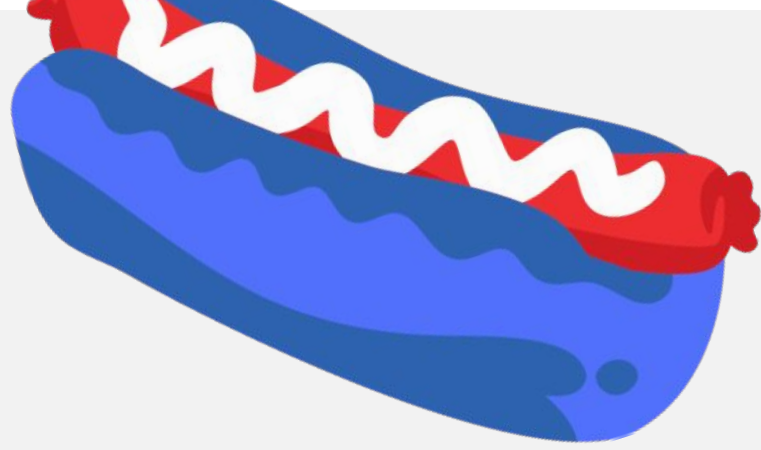
Results and Conclusions

- Total speeches analyzed: 171 (1850 - 2025)
- Mean % positive sentences:
 - Crisis years: 56.10%
 - Non-crisis years: 56.87%
- Two-sample t-test:
 - $t = -0.5828$
 - $p = 0.5608$ ($\alpha = 0.05$)
- Result: **No statistically significant** difference in sentiment between crisis and non-crisis speeches

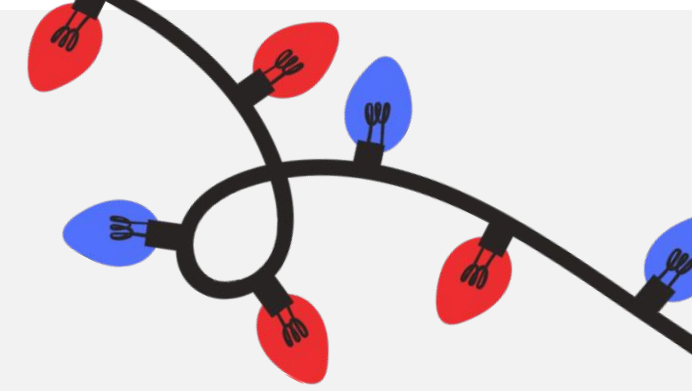


Hypothesis: During times of national crisis, more than 50% of State of the Union addresses given in the same year as the crisis will be classified as more optimistic.





Next Steps



New Lines of Exploration

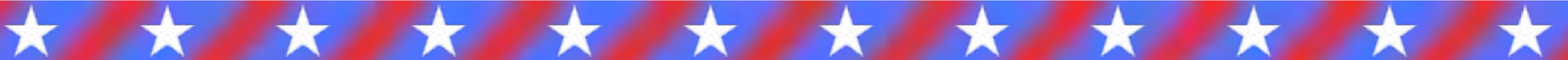
- Test whether presidential party affects sentiment.
- Conducting a **Topic Analysis** to examine how issue focus shifts during crisis vs. non-crisis years.

Improvements

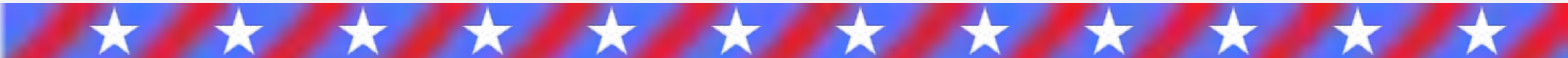
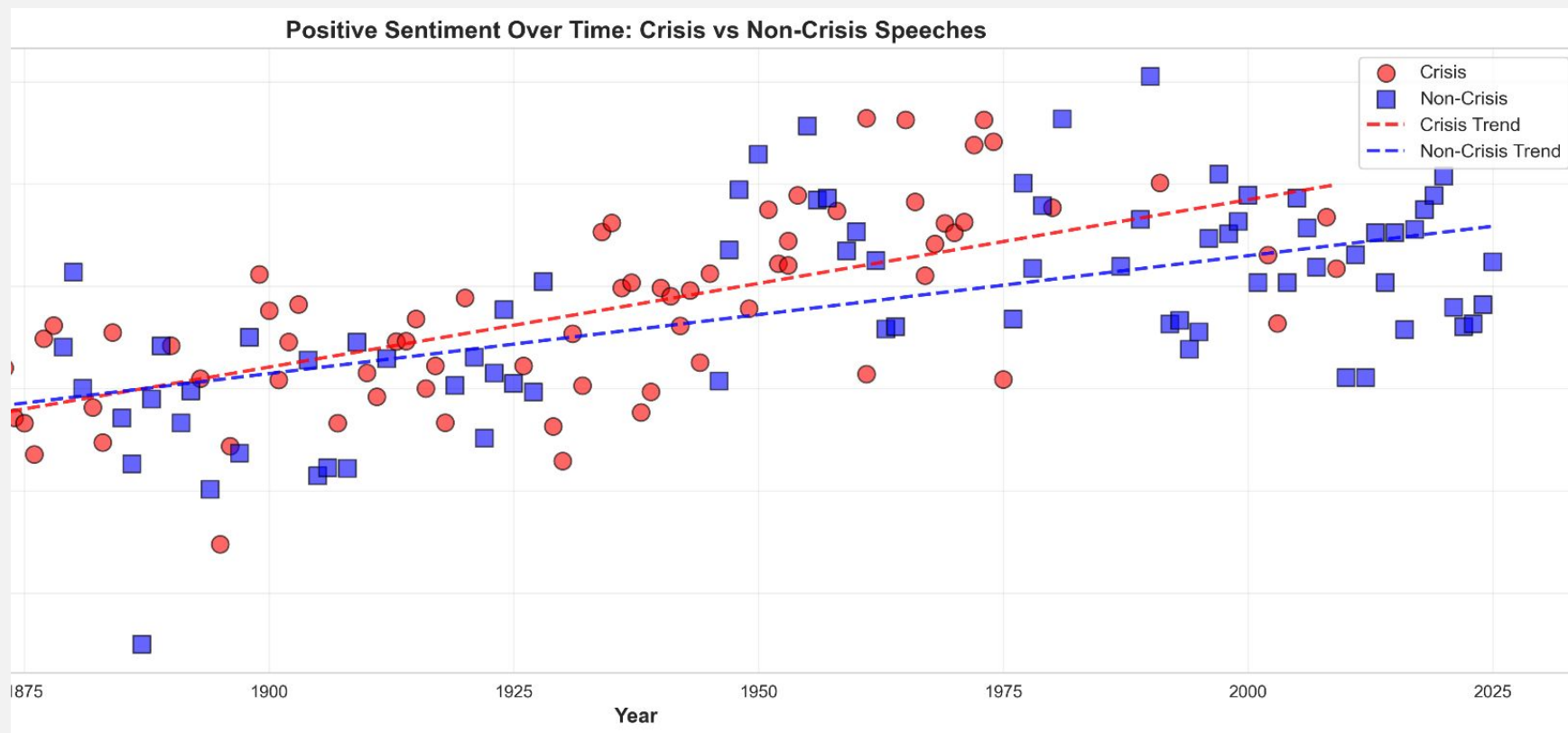
- Refine crisis classification
- Make the model more predictive, e.g. being able to classify whether we are in a period of crisis based solely on the speech.

New Questions

- Has political rhetoric become structurally more optimistic over time?
- Has rising political **polarization** affected the amount of optimism in SOTU speeches?

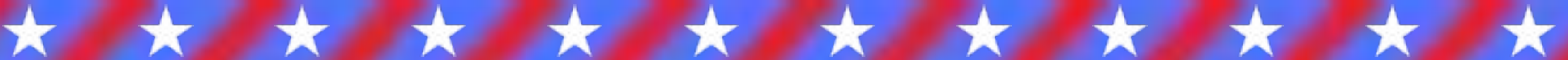


As the next SOTU approaches on Feb 26, will political rhetoric reflect crisis concerns, or continue the upward optimism trend?



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Appendix A: Full List of “Crisis period”

			Event	Start	End
Event	Start	End	Recession	August 1918	March 1919
			Recession	January 1920	July 1921
Mexican-American War	April 1846	February 1848	Recession	October 1926	November 1927
Panic of 1857	June 1857	December 1858	Great Depression	August 1929	1939
Civil War	April 12, 1861	May 26, 1865	Recession	May 1937	June 1938
Recession	April 1865	December 1867	World War II	September 1, 1939	September 2, 1945
Recession	June 1869	December 1870	Recession	November 1948	October 1949
Panic of 1873 / Long Depression	October 1873	March 1879	Korean War	June 25, 1950	July 27, 1953
Recession	March 1882	May 1885	Recession	July 1953	May 1954
Recession	July 1890	May 1891	Vietnam War	November 1, 1955	April 30, 1975
Panic of 1893	January 1893	June 1894	Recession	August 1957	April 1958
Panic of 1896	December 1895	June 1897	Recession	April 1960	February 1961
Spanish-American War	April 1898	August 1898	Cuban Missile Crisis	October 1962	October 1962
Philippine-American War	February 1899	July 1902	Recession	November 1973	March 1975
Recession	September 1902	August 1904	Recession	January 1980	July 1980
Panic of 1907	May 1907	June 1908	Recession	July 1981	November 1982
Panic of 1910–11	January 1910	January 1912	Early 90s Recession	July 1990	March 1991
Recession	January 1913	December 1914	Dot Com Recession	March 2001	November 2001
World War I	July 28, 1914	November 11, 1918	War on Terror	September 2001	May 2003
			Great Recession	December 2007	June 2009
			COVID-19 Pandemic / Recession	March 2020	May 2020